

Project Workflow Testing

1. Testing Objective

The purpose of this testing phase was to verify that the Volta Coffee wholesale order workflow behaves correctly from end to end.

The workflow was tested to confirm that it can:

- receive a wholesale order through the n8n form,
- extract structured information using Google Gemini,
- parse and validate the AI response,
- save the order in Airtable,
- classify the order correctly,
- route the order to the correct workflow branch,
- send the appropriate email response.

The tests covered three main business scenarios:

1. a complete standard order,
2. an order requiring human review,
3. an incomplete order requiring clarification.

These scenarios were selected because they represent the three possible classifications used in the workflow:

- standard
 - needs_review
 - needs_clarification
-

2. Test Environment

The workflow was tested using the following tools:

- n8n for workflow automation
- Google Gemini for information extraction and order classification
- JavaScript for parsing and validating the AI response
- Airtable for order storage
- Gmail for automated customer communication

The complete workflow was tested from the initial form submission to the final Airtable record and email output.

3. Test Cases

Test Case TC-01 – Standard Order

Test Objective

To confirm that a complete and unambiguous wholesale order is classified as standard and routed to the confirmation email branch.

Input

Hello,

We would like to order:

20 kg Ethiopia Sidamo

15 kg Brazil Santos

Please deliver on 2026-08-05 between 08:00 and 11:00.

Thank you.

Expected Result

- Gemini extracts the two coffee products and quantities.
- The delivery date is returned in ISO format.
- The delivery time is extracted correctly.
- The order is classified as standard.
- The structured order is saved in Airtable.
- The Switch node routes the order to the Standard branch.
- A confirmation email is sent.

Actual Result

- Classification returned: standard
- Items and quantities were extracted correctly.
- Delivery information was extracted.
- Order data was stored in Airtable.
- The Standard branch was executed.
- The confirmation email was received.

Classification Reason

The order contained clear products, quantities, delivery date, and delivery time. No substitution, ambiguity, or missing information was detected.

Result

Passed

Test Case TC-02 – Needs Review

Test Objective

To confirm that an order containing a substitution request is classified as needs_review because it requires a human decision.

Input

Hello,

Please send:

20 kg Ethiopia Sidamo

If it is unavailable, another washed Ethiopian coffee is acceptable.

Delivery on 2026-08-05.

Thank you.

Expected Result

- Gemini extracts the requested product and quantity.
- The substitution request is detected.
- The order is classified as needs_review.
- The order is saved in Airtable.
- The Switch node routes the order to the Needs Review branch.
- A review email is sent.

Actual Result

- Classification returned: needs_review
- The requested coffee and quantity were extracted.
- The substitution request was detected.
- The order was stored in Airtable.
- The Needs Review branch was executed.
- The review email was received.

Classification Reason

The customer allowed an alternative product if Ethiopia Sidamo was unavailable. This requires a human to verify availability and approve the substitution before the order can be confirmed.

Result

Passed

Test Case TC-03 – Needs Clarification

Test Objective

To confirm that an incomplete order is classified as needs_clarification and routed to the clarification email branch.

Input

Hello,

Please send some Colombia coffee.

Deliver next Tuesday.

Thanks.

Expected Result

- Gemini identifies Colombia coffee as the requested product.
- The missing quantity is detected.
- The relative delivery date is identified as potentially unclear.
- The order is classified as needs_clarification.
- The order is saved in Airtable.
- The Switch node routes the order to the Needs Clarification branch.
- A clarification email is sent.

Actual Result

- Classification returned: needs_clarification
- The product request was extracted.
- The missing quantity was identified.
- The delivery date was interpreted, while the missing information was recorded in the human notes.
- The order was stored in Airtable.
- The Needs Clarification branch was executed.
- The clarification email was received.

Classification Reason

The customer did not provide a quantity. The phrase “next Tuesday” also depended on the submission date and required careful interpretation. Additional information was therefore needed before processing the order.

Result

Passed

4. Validation Summary

Test ID	Scenario	Expected Classification	Actual Classification	Airtable Record	Email Triggered	Result
TC-01	Complete order	standard	standard	Yes	Confirmation	Passed
TC-02	Substitution request	needs_review	needs_review	Yes	Review	Passed
TC-03	Missing quantity and unclear date	needs_clarification	needs_clarification	Yes	Clarification	Passed

All three main classification paths were tested successfully.

5. Validation Checklist

Validation Check	Result
Form submission triggered the workflow	Passed
Customer information was preserved	Passed
Gemini returned structured output	Passed
JSON was parsed successfully	Passed
Classification value was standardized	Passed
Order was stored in Airtable	Passed
Correct Switch branch was selected	Passed
Appropriate email was sent	Passed
Missing information was recorded for human review	Passed
Workflow completed without unresolved errors	Passed

6. Bugs and Issues Identified During Development

Several issues were discovered while building and testing the workflow. These were corrected before the final validation.

Bug 1 – Invalid or Inconsistent JSON

Bug Type

AI Behaviour Bug

Problem

Gemini did not always return the output in a format that could be parsed reliably. Extra text or malformed JSON could cause the next node to fail.

Cause

The original prompt did not restrict the output format strongly enough.

Fix Applied

The Gemini prompt was updated to:

- return JSON only,
- use predefined field names,
- avoid explanatory text outside the JSON,
- use only the allowed classification values.

A JavaScript parser was also added to parse, clean, and normalize the response.

Retest Result

Passed

Bug 2 – Incorrect Date Format in Airtable

Bug Type

Data Flow Bug

Problem

Airtable rejected some delivery dates when they were not formatted correctly.

Cause

The AI sometimes returned dates in natural-language format rather than the ISO format expected by Airtable.

Fix Applied

The prompt was updated to require the following format:

YYYY-MM-DD

The output was then validated before being sent to Airtable.

Retest Result

Passed

Bug 3 – Empty Delivery Date Handling

Bug Type

Data Flow Bug

Problem

When the delivery date was unclear or unavailable, Airtable could reject the value or store it incorrectly.

Cause

An empty string was being sent to a date field.

Fix Applied

The Airtable mapping was changed to return a null value when no valid date was available:

```
{{ $json.delivery_date || null }}
```

Retest Result

Passed

Bug 4 – Classification Value Mismatch

Bug Type

Logic Bug

Problem

The Switch node depended on exact classification labels. A small difference in spelling, capitalization, or spacing could prevent the correct branch from running.

Cause

AI-generated text can be inconsistent if the output is not normalized.

Fix Applied

The JavaScript node was used to:

- convert the classification to lowercase,
- remove unnecessary spaces,
- restrict the value to the three approved classifications.

Approved values:

standard

needs_review

needs_clarification

Retest Result

Passed

Bug 5 – Gmail Credential Connection

Bug Type

Integration / Authentication Issue

Problem

The Gmail node stopped working because the Google OAuth connection required reconnection.

Cause

The credential session had expired or was no longer authenticated.

Fix Applied

The Gmail credential was reconnected in n8n and the email nodes were tested again.

Retest Result

Passed

7. Reliability and Safety Checks

The workflow includes several controls to reduce the risk of incorrect automated decisions.

Structured AI Output

Gemini is required to return a fixed JSON structure rather than free text. This makes the output easier to validate and use in later nodes.

Restricted Classification Values

The AI can only assign one of three classifications:

- standard
- needs_review
- needs_clarification

This prevents unexpected labels from reaching the Switch node.

JavaScript Validation

The parser checks and normalizes the AI output before it is stored or routed.

Human Review

Orders involving substitutions, ambiguity, or unusual requests are not treated as routine orders. They are classified as `needs_review` or `needs_clarification`.

Original Message Preservation

The original customer order text is stored in Airtable so that a human can compare the AI-extracted information with the original request.

No Invented Information

The Gemini prompt instructs the model not to invent missing quantities, dates, products, or customer information.

8. Testing Limitations

The final tests covered the three main workflow branches, but the project does not currently include:

- live inventory checking,
- automatic product availability validation,
- price calculation,
- invoice generation,
- customer identity verification,
- advanced email-format validation,
- multilingual testing across several languages.

These features are outside the current project scope but could be added in a future version.

9. Test Evidence

The following screenshots were added as evidence:

1. Customer order form submission
2. Airtable records created by the workflow
3. Standard order confirmation email
4. Needs Review email
5. Needs Clarification email
6. Final n8n workflow overview

The screenshots confirm that the customer input was processed, stored, classified, and routed to the appropriate email branch.

10. Final Testing Conclusion

The final validation confirmed that the workflow correctly handled the three main wholesale order scenarios.

For each test case:

- the form triggered successfully,
- Gemini extracted and classified the order,
- the JavaScript node parsed and normalized the output,
- Airtable stored the structured data,
- the Switch node selected the correct route,
- and the appropriate customer email was sent.

All three final test cases passed.

The development process also identified several AI behaviour, data flow, logic, and authentication issues. These were corrected and successfully retested.

The workflow is therefore considered functional and suitable for the scope of this project. Human review remains part of the process for orders that contain missing information, substitutions, or other conditions that should not be confirmed automatically.

APPENDIX A

Testing Evidence and Screenshots

Volta Coffee Wholesale Order Processing Workflow

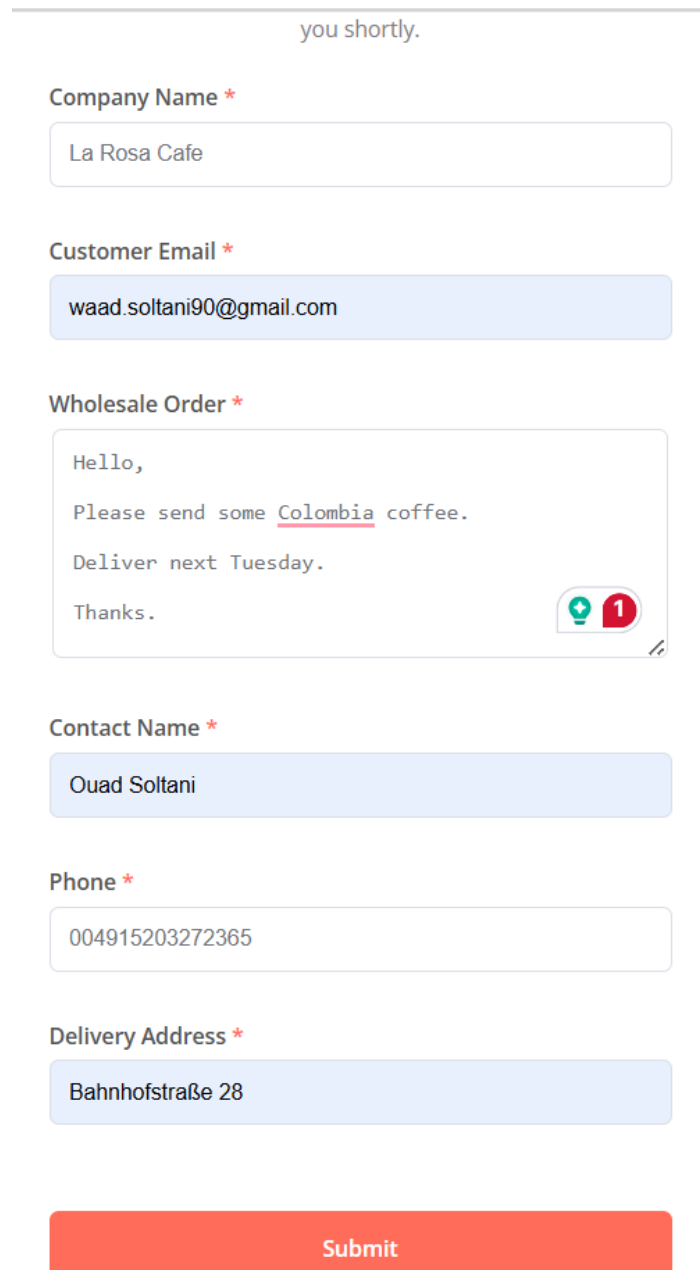
This appendix provides visual evidence from the final workflow testing. The screenshots show the customer input form, the complete n8n workflow, the Airtable testing records, and customer email outputs for the Standard and Needs Clarification routes.

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A.1 Customer Order Input

The form below shows a Needs Clarification test input. The customer requested "some Colombia coffee" without specifying a quantity and used the relative date "next Tuesday".



The screenshot shows a web form for a customer order. At the top, there is a greyed-out text "you shortly." Below this are several input fields, each with a red asterisk indicating a required field. The fields are: "Company Name" with the value "La Rosa Cafe"; "Customer Email" with the value "waad.soltani90@gmail.com"; "Wholesale Order" with a text area containing "Hello, Please send some Colombia coffee. Deliver next Tuesday. Thanks." and a small icon of a lightbulb with a red circle containing the number 1; "Contact Name" with the value "Ouad Soltani"; "Phone" with the value "004915203272365"; and "Delivery Address" with the value "Bahnhofstraße 28". At the bottom of the form is a large red "Submit" button.

you shortly.

Company Name *

La Rosa Cafe

Customer Email *

waad.soltani90@gmail.com

Wholesale Order *

Hello,
Please send some Colombia coffee.
Deliver next Tuesday.
Thanks.

Contact Name *

Ouad Soltani

Phone *

004915203272365

Delivery Address *

Bahnhofstraße 28

Submit

Figure A1. Customer order form used for the Needs Clarification test case.

Evidence for TC-03: incomplete quantity information and a relative delivery date.

A.2 Complete n8n Workflow

The workflow processes each order from intake through AI extraction, JSON validation, Airtable storage, classification-based routing, and automated customer communication.

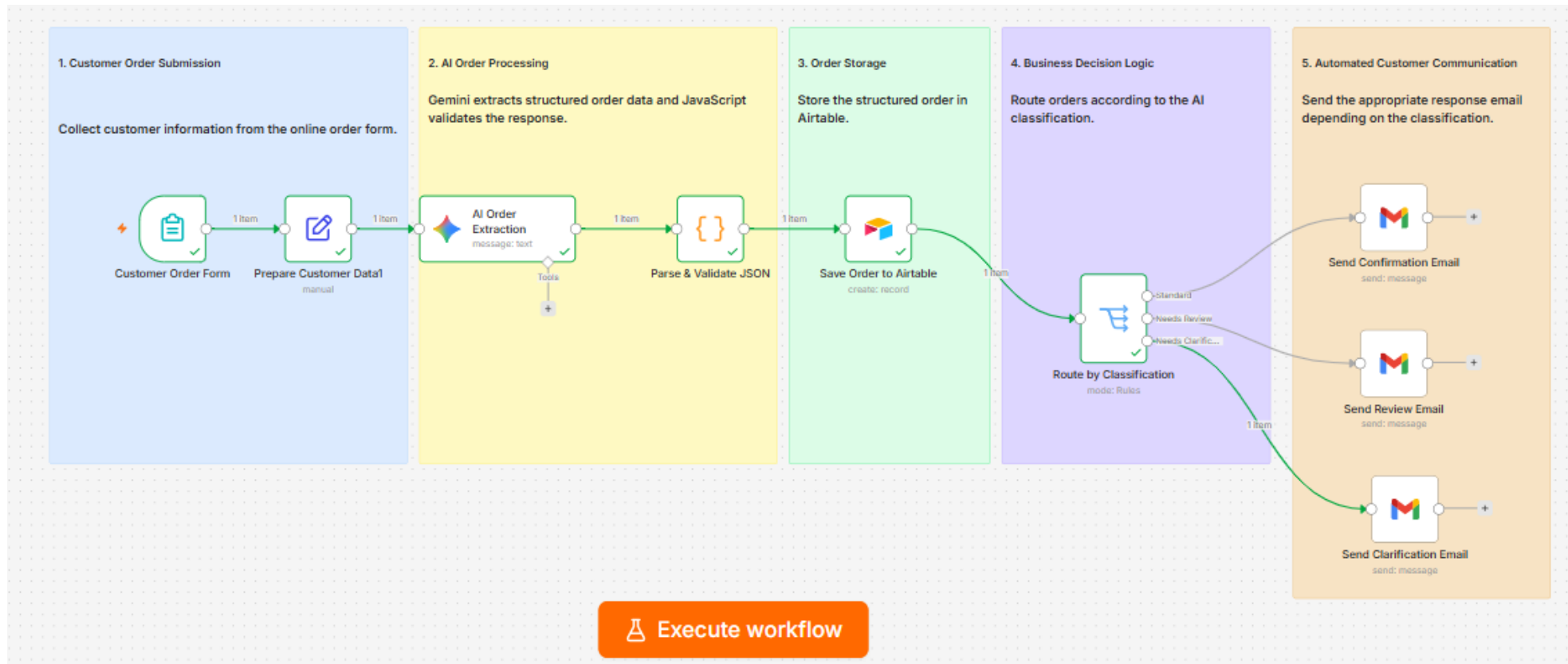


Figure A2. Final end-to-end n8n workflow.

The three routes are Standard, Needs Review, and Needs Clarification.

A.3 Airtable Testing Records

The Airtable base contains structured order records created during testing. The classification field shows examples of all three workflow outcomes.

	company_name	delivery_address	delivery_date	items	classification	notes_for_human	raw_order_text	delivery_time
1	Sunrise Café	Taubenbogen 5	27/7/2026	20 kg - Ethiopia Sidamo (W...	needs_clarification	The delivery date is contra...	Hello Volta Coffee, We wou...	8:00 - 11:00 AM
2	Sunrise Café	Taubenbogen 5	5/8/2026	20 kg - Ethiopia Sidamo (W...	needs_review	Customer requested a subs...	Hello Volta Coffee, We wou...	between 8:00 and 11:00 AM
3	Blue Horizon Café	12 Coffee Street, Berlin	6/8/2026	15 kg - Ethiopia Sidamo (W...	standard		Hello Volta Coffee, We wou...	9:00 AM and 11:00 AM
4	Blue Horizon Café	12 Coffee Street, Berlin	6/8/2026	15 kg - Ethiopia Sidamo (W...	standard		Hello Volta Coffee, We wou...	between 9:00 AM and 11:0...
5	Cafe Sola	Bahnhofstraße 28			needs_clarification	The customer has not speci...	Example: Send us some cof...	
6	La Rosa Cafe	Bahnhofstraße 28	4/8/2026	0 kg - Colombia coffee	needs_clarification	The quantity of coffee was ...	Hello, Please send some Co...	
+								

Figure A3. Airtable records generated during workflow testing.

Visible classifications include standard, needs_review, and needs_clarification.

A.4 Needs Clarification Email

This email was triggered after the workflow identified missing quantity information. The message explains the issue and asks the customer to provide the required details.

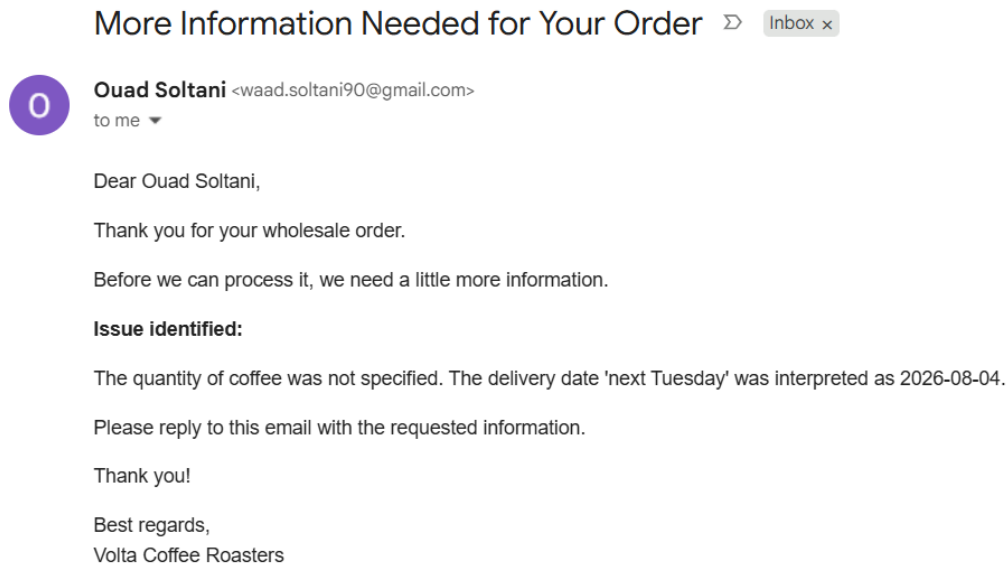


Figure A4. Customer email generated for a Needs Clarification order.

Evidence for TC-03: the missing quantity was identified and communicated.

A.5 Standard Order Confirmation Email

This confirmation email was generated for a complete order that was classified as Standard. It includes the customer name, extracted order items, and delivery information.

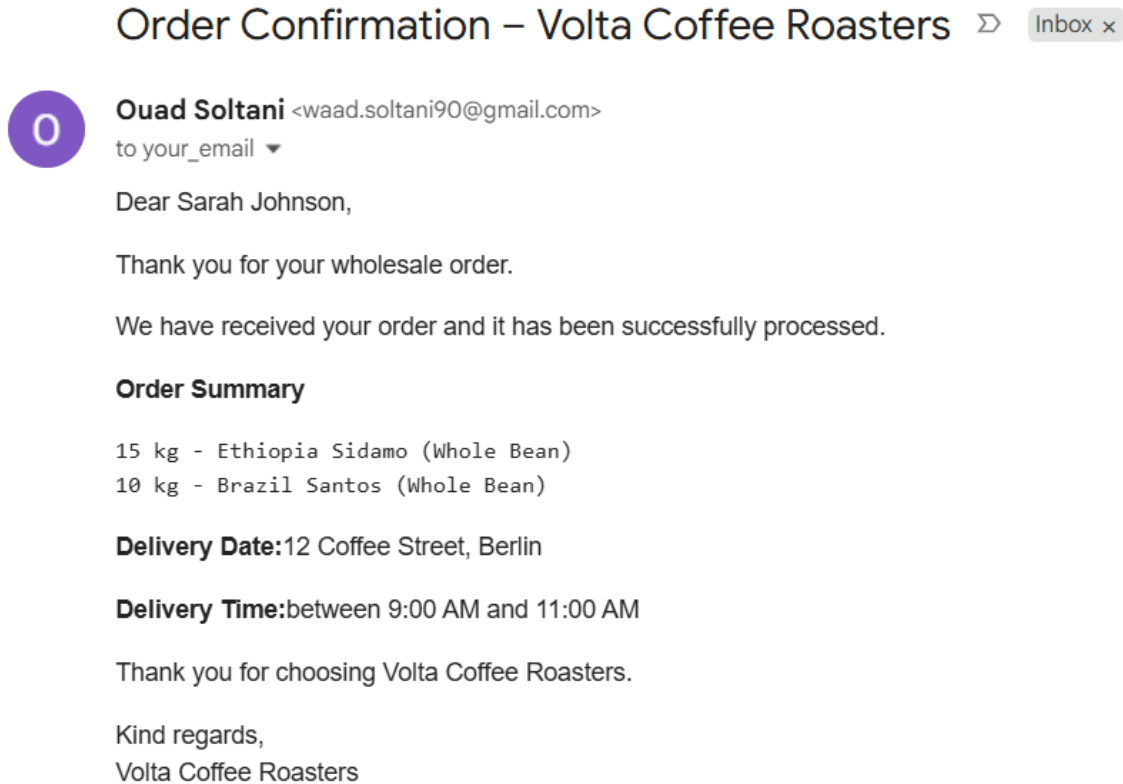


Figure A5. Customer confirmation email generated for a Standard order.

Evidence for TC-01: the order followed the Standard branch successfully.

Evidence note: The uploaded screenshot set does not contain a separate image of the Needs Review email. The Needs Review classification is nevertheless visible in the Airtable evidence in Figure A3.